

Customer Segmentation and Targeted Marketing - Executive Report

Course: Machine Learning II, NOVA IMS Datasets: customer_info (about 33,000 customers) and customer_basket (100,000 baskets)

The code for this work is in Code.py and final_project.ipynb. This document describes the results and the proposed campaigns; the figures below are generated by that code.

1. Executive Summary

We grouped about 33,000 customers into 5 segments using unsupervised learning, and then used their shopping baskets to design a promotion for each segment.

We cleaned and scaled the data and tried three clustering methods: K-Means, Self-Organizing Maps (SOM) and hierarchical clustering. The elbow and silhouette analysis both pointed to 5 clusters. We kept SOM as the final model because it orders the segments along a smooth gradient that is easy to explain. Hierarchical clustering was used only to double-check the structure.

The five segments are: Breakfast Lovers (large families), Adult Gift Buyers (tech), Families with Pets, Healthy and Vegans, and Tech and Athletes.

For each segment we mined association rules (Apriori) on the baskets and turned the strongest ones into concrete offers, for example "buy eggs and butter, get 20% off fresh bread", or an AirPods coupon when a phone is added to the cart.

Finally, we placed the segments on a profitability vs. stability chart to see where the value and the risk are. The most valuable and reliable segments are the Adult Gift Buyers and the Breakfast Lovers. The Tech and Athletes segment brings good revenue but is the least loyal and complains the most, so it needs the most attention on retention.

2. Exploratory Data Analysis and Pre-Processing

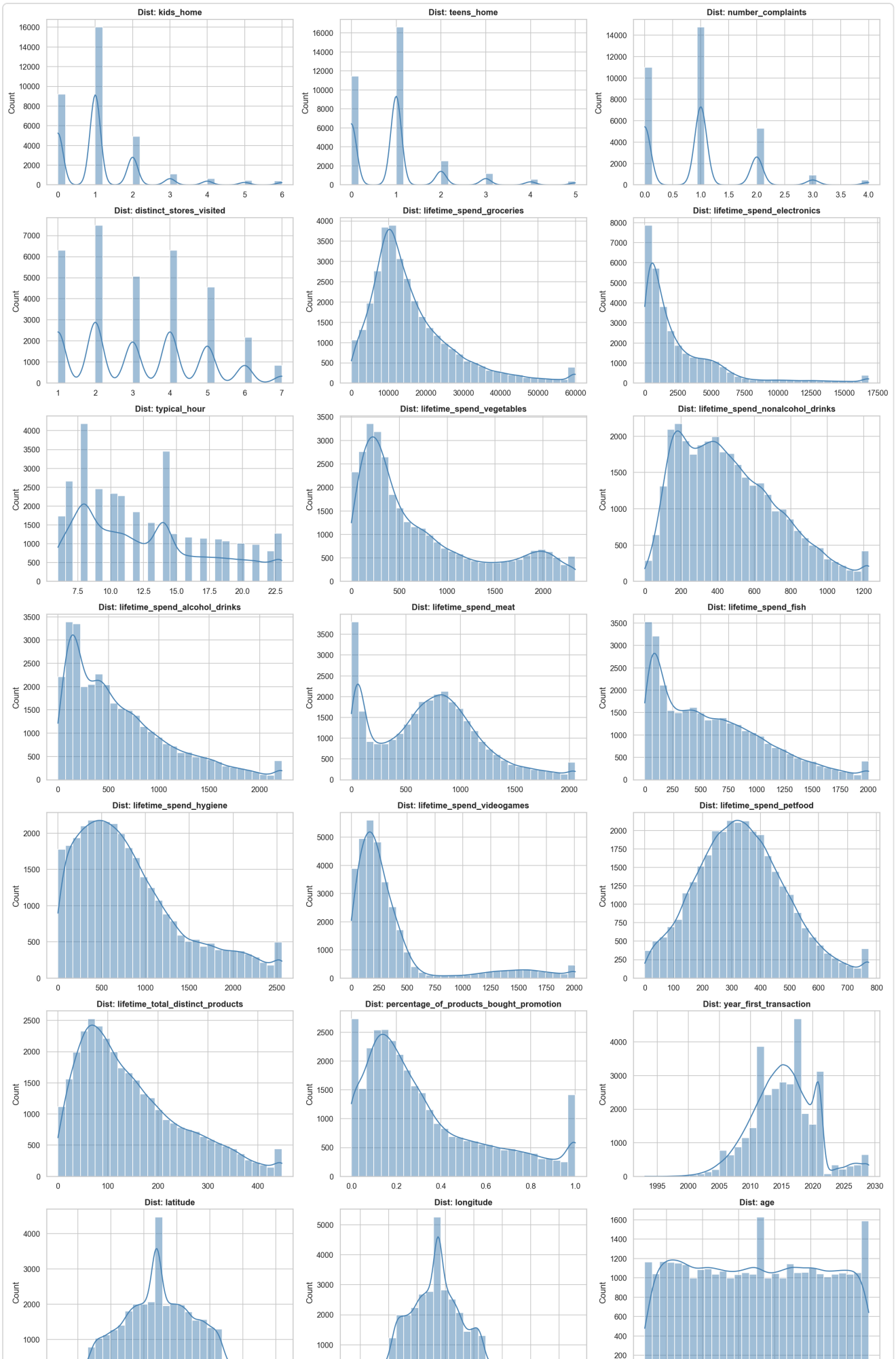
Feature engineering:

- We turned customer_birthdate into age so it can be used in distance-based clustering.
- We turned loyalty_card_number into has_loyalty_card (1/0). The card numbers are only identifiers, so keeping them as numbers would distort the distances.
- We moved customer_id to the index and dropped the name and raw id columns.

Data cleaning:

- Sanity checks: spend and count columns clipped at 0, promotion share capped at 100%, typical_hour capped at 23, and age limited to the 14 to 100 range.
- Outliers: the financial columns were capped at the 99th percentile so that a few very large spenders do not pull the centroids.
- Missing values: age was filled with the median; the other numeric columns were filled with a KNN imputer (k=5), which keeps the relationships between features better than filling with the global mean.

- Scaling: all features were standardized with StandardScaler so that no single large column dominates the distance.



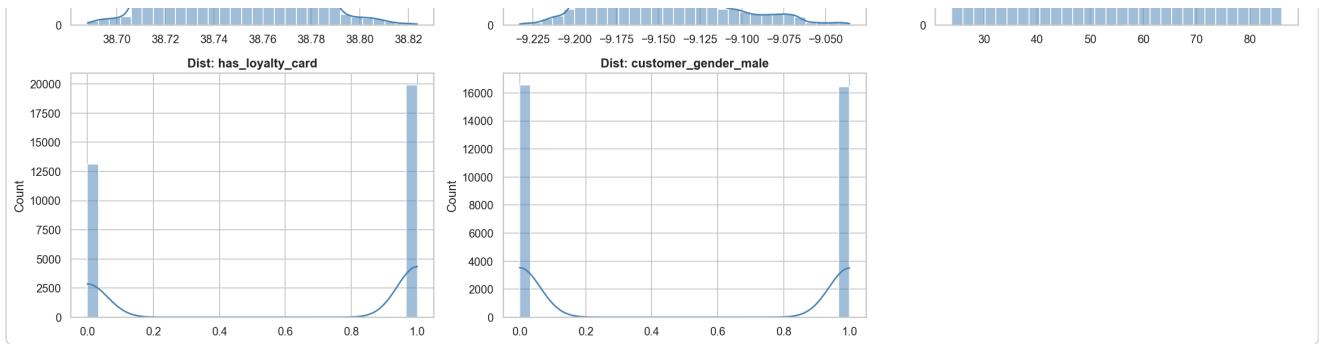


Figure 1: histograms of the numeric features.

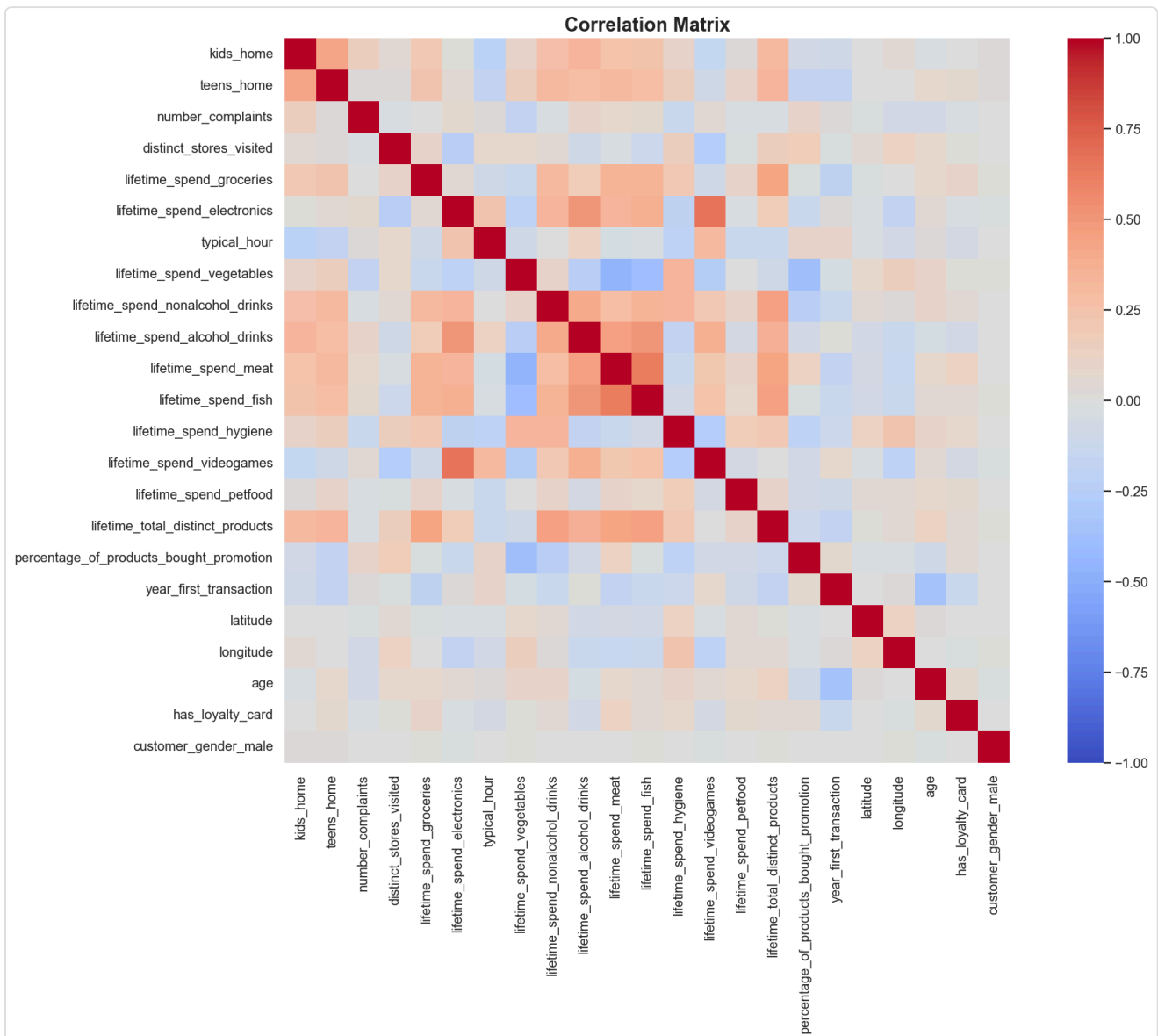


Figure 2: correlation heatmap, used to check for near-duplicate features.

After cleaning, the modelling table has 33,038 customers and 23 features, with no missing values.

3. Customer Segmentation and Clustering

Number of clusters. We computed the elbow (WCSS) and the silhouette score for k from 2 to 14. Both point to k=5, which also gives the segments that make the most business sense.

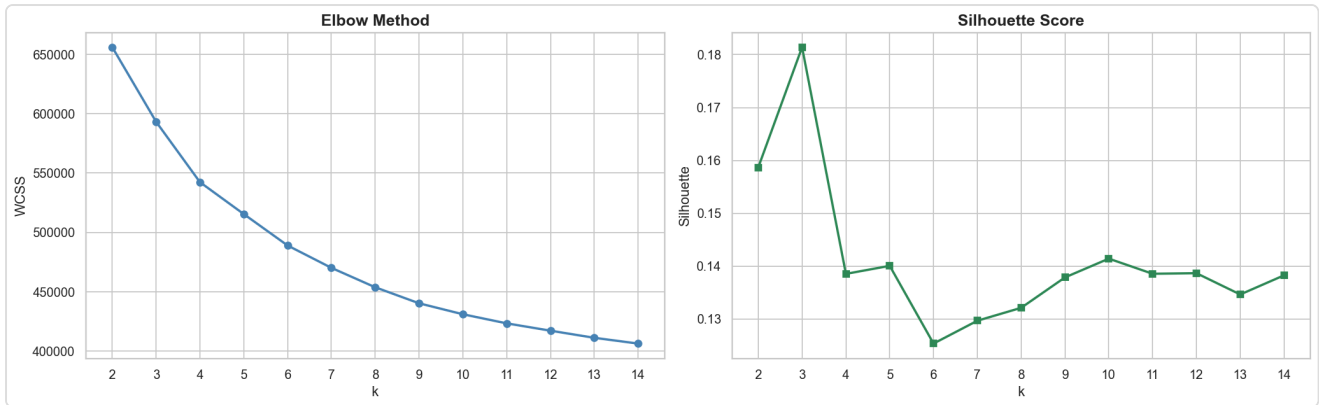


Figure 3: elbow and silhouette curves.

Models. We compared three different methods:

- K-Means: a fast baseline, good for compact clusters.
- SOM (5x1): the final model. It places the segments on a 1-D gradient where neighbouring clusters are genuinely similar, which makes them easier to describe.
- Hierarchical (agglomerative): used as a benchmark and for the dendrogram.

Hierarchical clustering and the choice of linkage. Hierarchical clustering needs $O(n^2)$ memory, and a dendrogram with 33,000 leaves is unreadable, so we ran the linkage comparison and the dendrograms on a random sample of 5,000 customers and then refit the chosen ward solution on the full data. The metrics on the sample were:

Linkage	Silhouette	Calinski-Harabasz	Davies-Bouldin
ward	0.130	556.2	2.21
complete	0.101	409.1	3.08
average	0.139	141.1	1.31

Average linkage has a slightly higher silhouette but a very low Calinski-Harabasz score. This is the usual chaining effect, where it produces one big cluster and a few tiny ones. Ward gives the most balanced and cohesive clusters (the highest Calinski-Harabasz and sensible group sizes), so we used ward as the representative hierarchical solution.

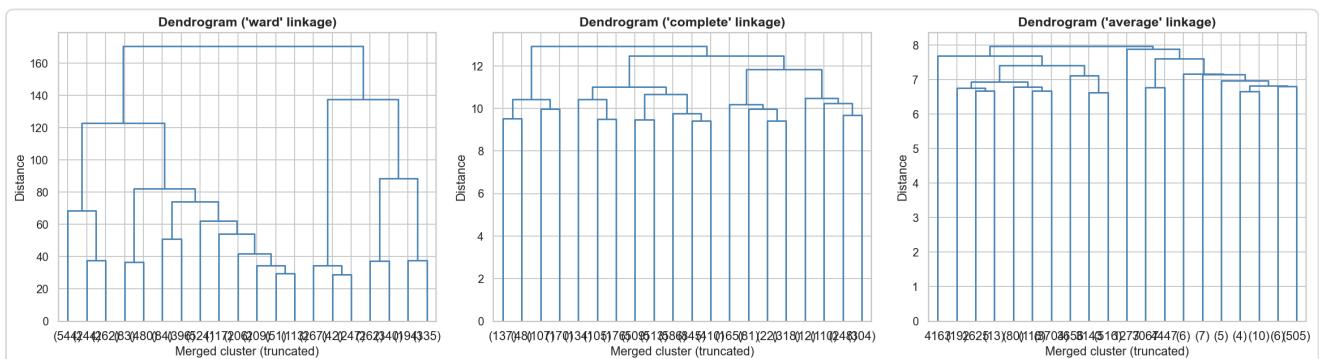


Figure 4: dendrograms for ward, complete and average linkage.

On the full data the three methods give similar internal scores (silhouette around 0.12 to 0.14, which is normal for this kind of high-dimensional behavioural data). K-Means and ward agree on the overall structure, which tells us the segmentation is stable and not an artefact of one method. We kept SOM as the final model for the reason given above.

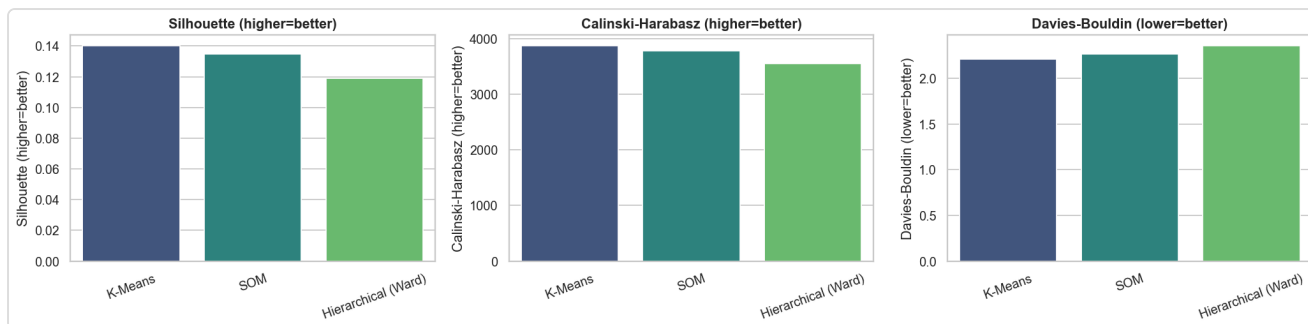
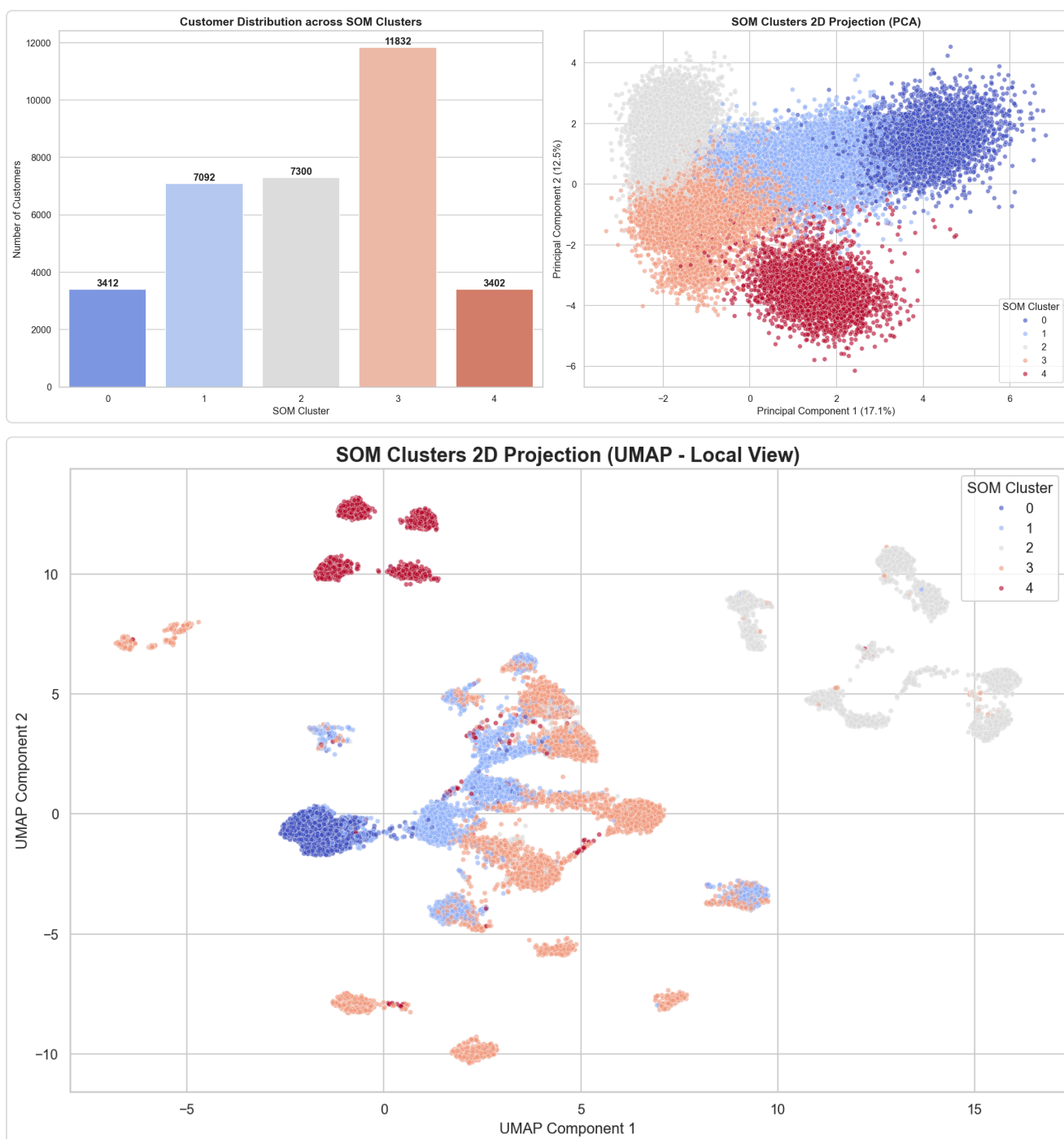


Figure 5: model comparison on silhouette, Calinski-Harabasz and Davies-Bouldin.

We also projected the data into 2-D with PCA and UMAP (local view with 15 neighbours, global view with 50) to confirm that the SOM segments are separable.



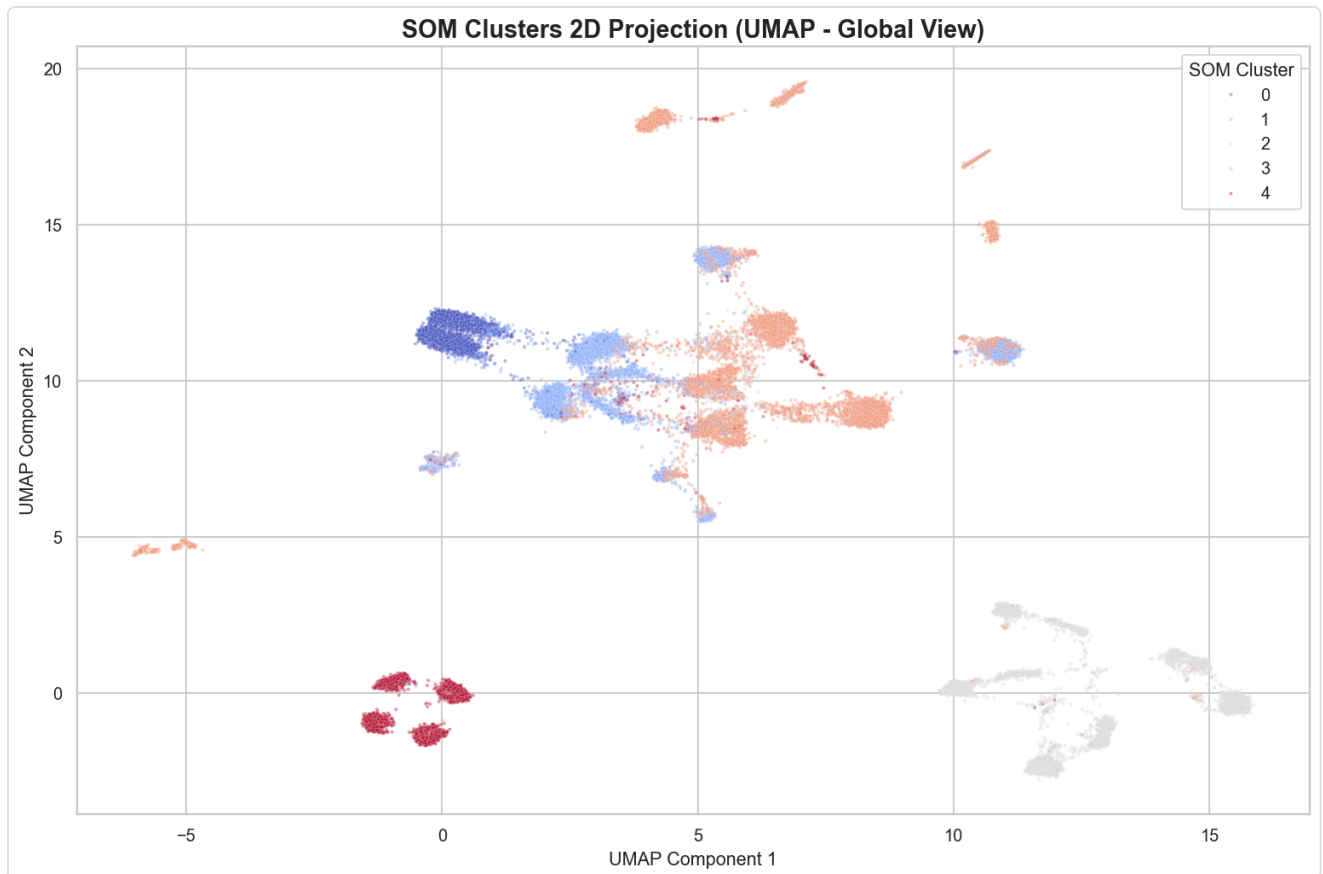


Figure 6: PCA and UMAP (local and global views) projections coloured by SOM cluster.

Segment profiles. The radar, the spend heatmap and the demographic bars make the segments concrete.

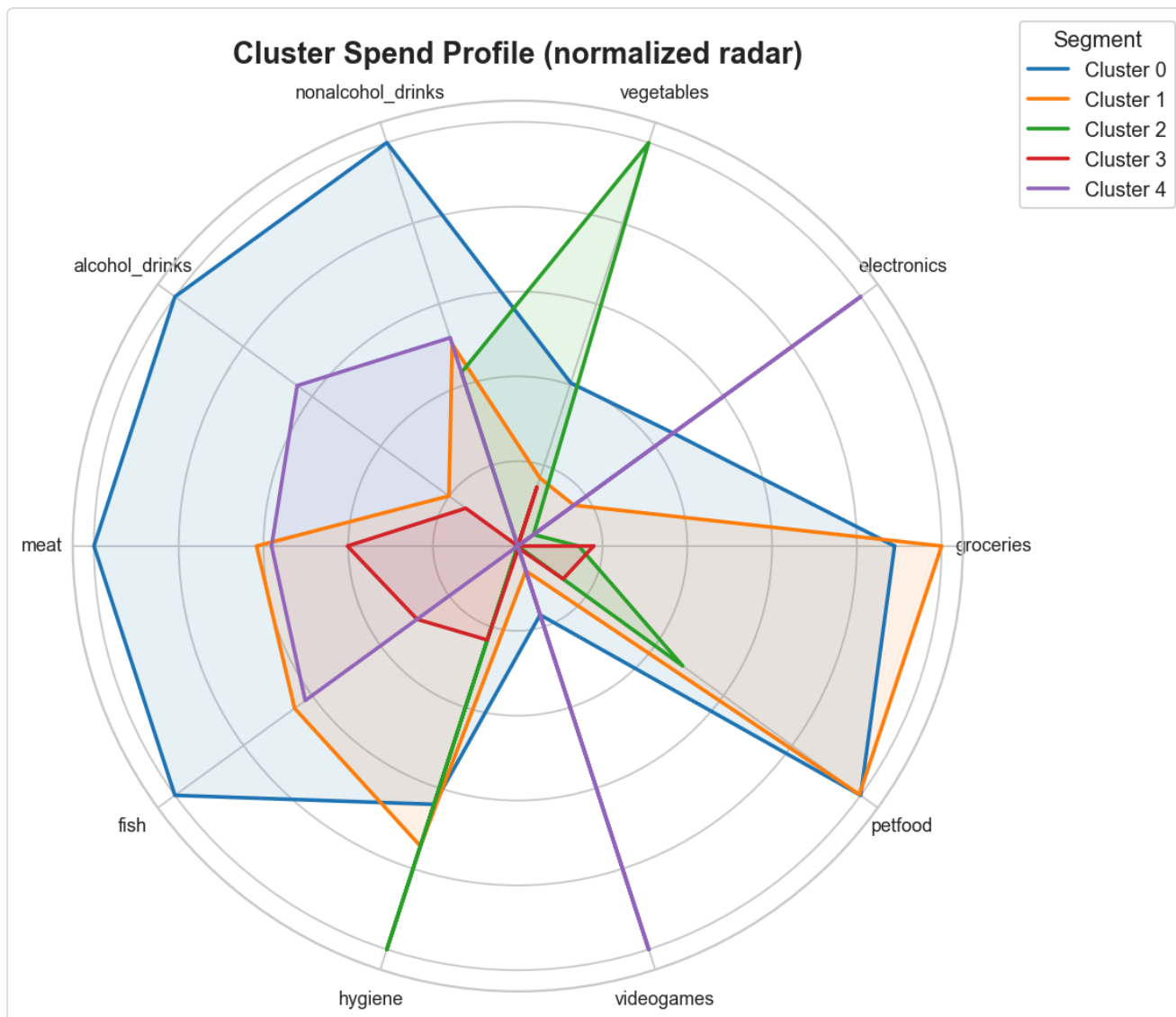


Figure 7: spend radar per cluster.

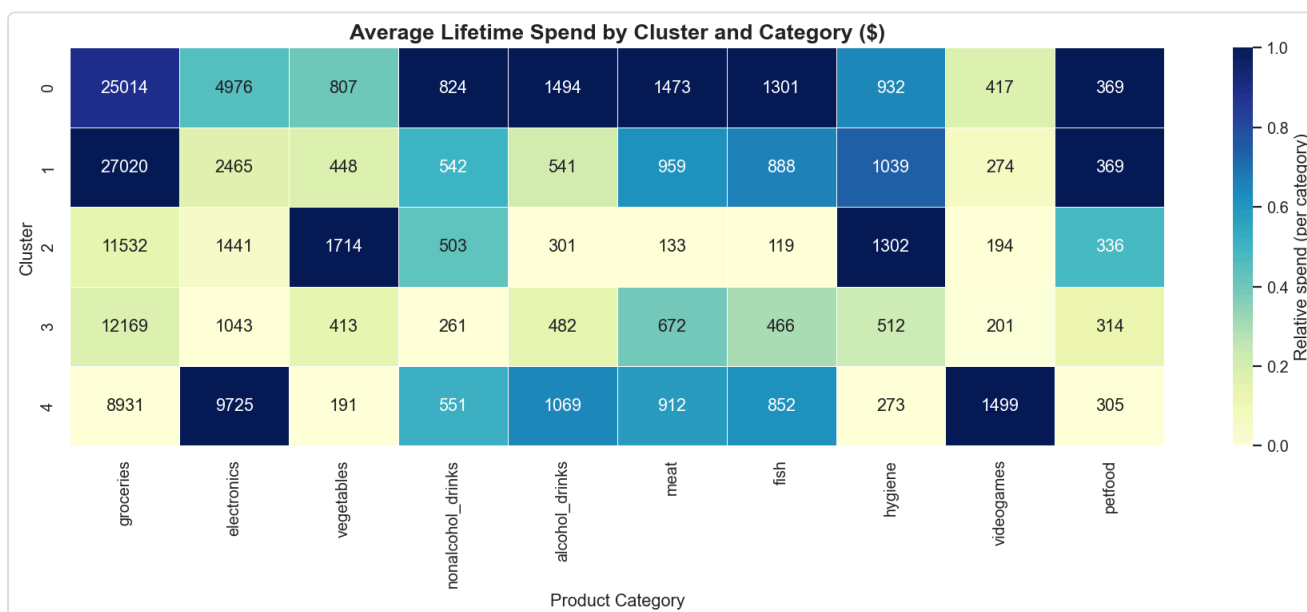


Figure 8: heatmap of average spend by cluster and category.

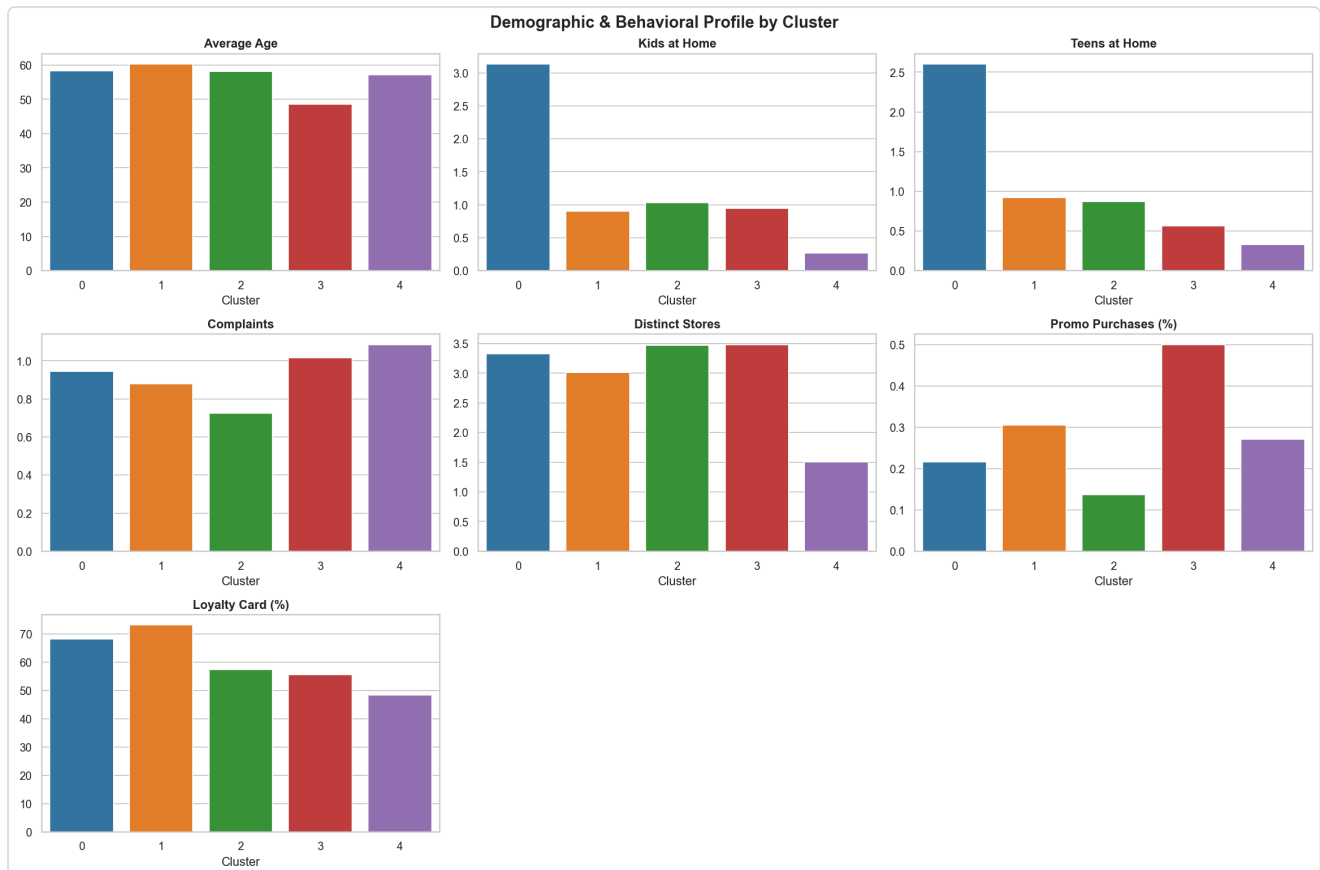


Figure 9: demographic and behavioural bars per cluster.

Cluster	Name	Age	Loyalty	Peak hour	LTV	Stability
0	Breakfast Lovers (large families)	58.3	68.2%	10:00	\$37.4k	78.6
1	Adult Gift Buyers (tech)	60.3	73.1%	11:00	\$34.4k	84.2
2	Families with Pets	58.2	57.4%	12:00	\$17.5k	70.1
3	Healthy and Vegans	48.6	55.6%	12:00	\$16.5k	65.2
4	Tech and Athletes	57.2	48.4%	17:00	\$24.0k	57.4

- Cluster 0, Breakfast Lovers: the largest households (about 3 kids on average), high loyalty, shop in the morning, and the highest lifetime value of all segments.
- Cluster 1, Adult Gift Buyers: the highest loyalty and lifetime value, the most reliable customers.
- Cluster 2, Families with Pets: the fewest complaints, and they mostly pay full price (only about 14% of products on promotion), so they are a good upsell target.
- Cluster 3, Healthy and Vegans: the youngest group, the lowest lifetime value and very price sensitive (about half of their products are bought on promotion).
- Cluster 4, Tech and Athletes: shop in the evening, visit very few stores (about 1.5), the lowest loyalty and the highest number of complaints, so they have the highest churn risk.

The Cluster 1 puzzle. The baskets of Cluster 1 (headphones, energy drinks, AirPods) looked like a young segment, but the average age is about 60. Looking closer, around 75% of these customers have teens or kids at home, and the electronics spend has rare but very large one-off purchases (over \$10,000). So this is an older group buying expensive tech as gifts for younger relatives. The messaging for this segment should be about gifts, not about a youth lifestyle.

Profitability vs. stability. We built a simple stability score (loyalty, minus half the promo share, minus the complaints) and plotted each segment by lifetime value against this score:

- Clusters 0 and 1 are the stars: high value and high stability, and they do not depend much on promotions.
- Cluster 2 is a cash cow: very stable with a moderate value, good for upsell and subscriptions.
- Clusters 3 and 4 are the riskier groups: lower stability, more complaints and more promo-driven buying.

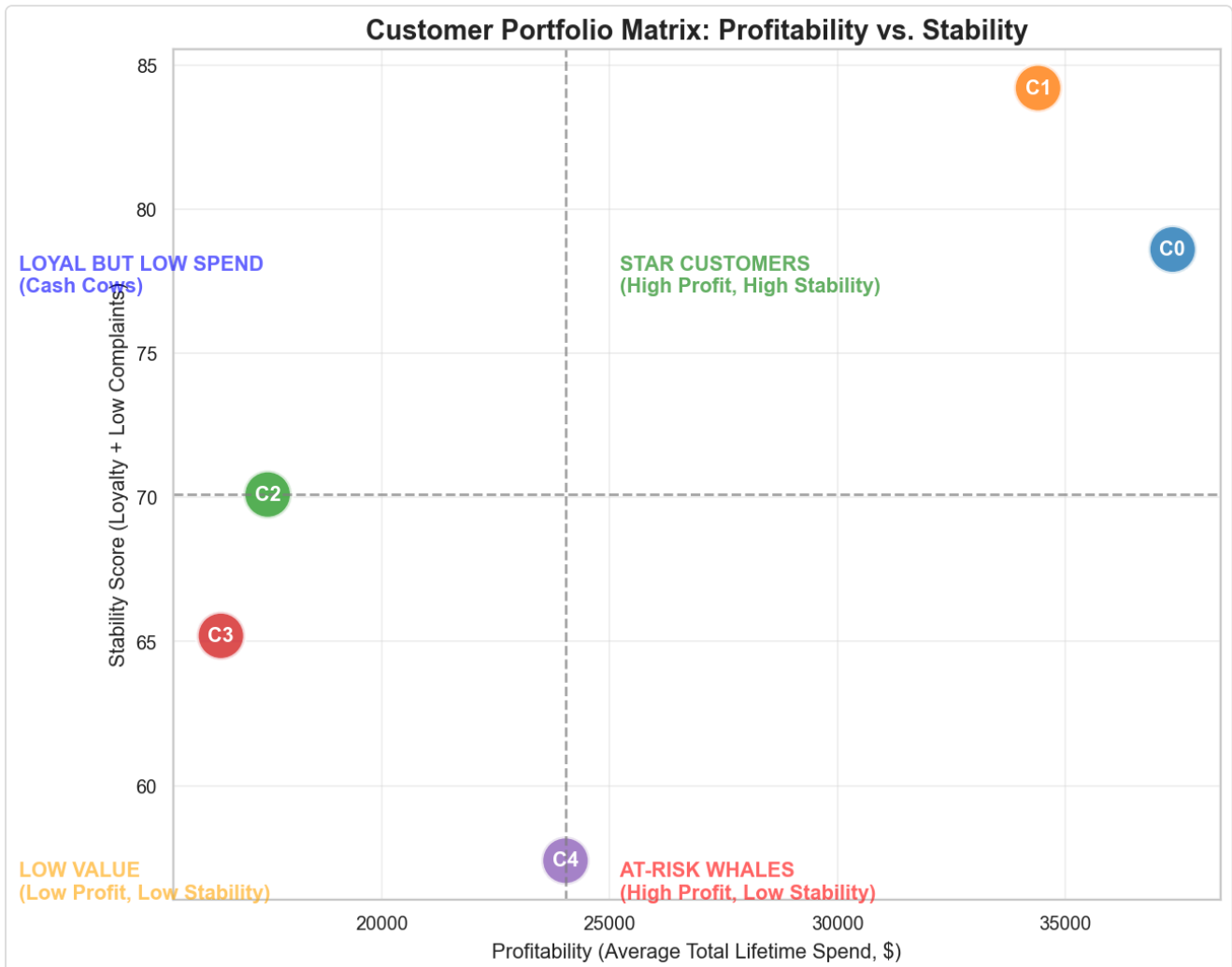


Figure 10: profitability vs. stability bubble chart.

4. Targeted Promotions

The offers come from Apriori association rules mined per segment, with $\text{min_support} = 0.02$, $\text{min_confidence} = 0.20$ and $\text{lift} > 1.2$. These thresholds drop trivial rules like "bread then milk" but keep the smaller, useful patterns.

Cluster 0, Breakfast Lovers (113 rules):

- Rules: butter + oatmeal then milk (lift 1.32); fresh bread + honey then tea.
- Offer: "buy eggs and butter, get 20% off fresh bread".
- Timing: send the email with breakfast recipes and a promo code at 09:00, one hour before they usually shop.

Cluster 1, Adult Gift Buyers (2 strong rules):

- Rules: energy drinks then AirPods (lift 2.16); Bluetooth headphones then AirPods.
- Offer: a "tech gifts" campaign that frames the electronics as gifts for children or grandchildren.
- These are star customers, so instead of direct discounts give them added value, for example a free pack of energy drinks with a headphone purchase.

Cluster 2, Families with Pets (40 rules):

- Rules: baby food then paper napkins (lift 2.07); dog food then napkins (lift 1.96).
- Offer: a "family basket": buy baby food and dog food, get a large pack of napkins at 30% off.
- Add a restock reminder for baby food and pet food every 2 to 3 weeks, with a small bonus for auto-pay.

Cluster 3, Healthy and Vegans (102 rules):

- Rules: avocado + asparagus then salad (confidence about 40%, lift 3.92).
- Offer: 20% off avocados when buying fresh asparagus and spinach, but only after the customer registers in the loyalty app. This group has low loyalty, so use the discount to bring them into the program.
- Send newsletters about fresh vegetables, and do not push meat.

Cluster 4, Tech and Athletes (1,375 rules):

- Rules: iPhone 10 then AirPods (lift 3.34); protein bar + deodorant then shampoo (lift 1.74).
- Offer: when a phone is added to the cart, automatically offer an AirPods coupon.
- A "post-workout" bundle (shampoo, deodorant, protein bar).
- Because this group complains the most, send an apology promo code automatically after negative feedback.

5. Conclusion and Recommendations

1. The five segments are stable across K-Means, SOM and ward, so we can trust them.
2. Spend the retention budget on the riskier groups (Tech and Athletes, Healthy and Vegans), and keep the star segments happy with loyalty rewards rather than discounts.
3. Upsell the Families with Pets segment, since they pay full price and rarely complain.
4. Change the message for Cluster 1 to focus on gifting.
5. Roll out the offers as A/B tests and track conversion rate, average order value and loyalty sign-ups.

Deliverables:

- Code.py: the data pipeline (preprocessing, clustering, comparison, recommendations).
- final_project.ipynb: the full analysis with the figures.
- customer_cluster_assignments.csv: every customer_id with its final cluster.
- This report.

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